

Weaver Goldman

 github.com/We-Gold  weavergoldman.com  linkedin.com/in/weaver-goldman  we.goldm@gmail.com

EDUCATION

Worcester Polytechnic Institute

Graduating May 2027

MS Data Science

GPA 4.0

BS Computer Science

GPA 4.0

Courses: MLOps, Deep Learning, NLP, Machine Learning, Statistical Methods for DS/ML, Database Management Systems, Business Applications in ML, Graph ML, Data Visualization

EXPERIENCE

AI/ML Engineering Intern | Regeneron Pharmaceuticals, Inc.

May 2026 – Aug 2026

- Built a platform to monitor the health of 30+ production AI models in one place, adopted by 3 pilot teams.
- Built ML-based drift detection that alerts teams when live data diverges from what a model was trained on, and tracked how data subgroups shift over time to help pinpoint the cause.
- Created a dashboard giving teams a single view of model performance, data drift, and benchmark results.
- Cut LLM evaluation costs by **over 60%** at equal confidence by tailoring questions to each model's past answers.

AI Engineering Intern | Medtronic (Minimally Invasive Therapies Group)

Jun 2025 – Aug 2025

- Saved **400+** hours of manual work annually by automating surgical robot log analysis tasks with AI.
- Built an AI assistant that analyzes surgical robot logs at an expert level in minutes instead of tens of hours.
- Connected the assistant to internal data sources with MCP and hybrid RAG to answer multi-step questions.

Web Development Lab Assistant | Laboratory of Spaceflight and Planetary Exploration

Aug 2024 – Present

- Under contract with **NASA JPL**, built a mission planning web app for exploring interplanetary trajectories.
- Let mission planners filter **millions** of trajectories interactively by optimizing the database and graphs.

Software Development and Research Intern | Gittlen Cancer Research Labs

May 2024 – Aug 2024

- Built a desktop app that lets researchers analyze large 3D medical scans on a laptop by extracting key structures.
- Designed a Docker-based plugin system so other labs can add their own segmentation methods to the tool.

RESEARCH EXPERIENCE

Ouroboros: A Tool for Segmentation of Complex Paths in Large 3D Datasets

May 2024 – Present

D. Northover, **W. Goldman**, D. Vanselow, K.C. Cheng. *Manuscript in preparation.*

Penn State College of Medicine

- Created the core algorithms for a tool that isolates and segments nerves and vessels in **terabyte-scale** 3D scans.
- Shrank the data researchers handle by **over 99%**, processing a **2.8 TB** scan in 6 minutes on a laptop.

Scaling Diffusion Models to Large Sparse Graphs

Aug 2025 – May 2026

D. Dechantsreiter, **W. Goldman**, B. Hu, H. Pham. *WPI MQP Report.*

Advised by F. Murai & O. Mangoubi

- Led development of two AI models that generate realistic networks at scales existing methods can't handle.
- Trained on networks where all prior methods ran out of memory, and made an existing model's generation step up to **116×** faster with **89×** less memory.

PROJECTS

#TeamGraduate AI | Python, FastAPI, React, WordPress, Docker, Google Cloud Run

Jan 2026 – May 2026

- Built AI tools that turn teachers' lessons into flashcards and study guides aligned to Namibia's curriculum.
- Reduced lesson creation time by **87.7%** with AI authoring tools built into the existing website.

Identifying Transit Accessibility Gaps in NYC | Scikit-Learn, Python, Plotly, GTFS

Apr 2025

- Found underserved NYC neighborhoods and forecast revenue for potential new stations using **6** years of data.
- Pinpointed potential stops predicted to earn over **\$4 million** annually, using interactive opportunity maps.

SKILLS

Languages & Tools: Python, SQL, JavaScript/TypeScript, C/C++, Java, L^AT_EX, Bash, Git, Linux, Figma, Jira

ML & Data Science: PyTorch, scikit-learn, TensorFlow, Pandas, NumPy

AI & LLMs: LangChain/LangGraph, RAG, Model Context Protocol (MCP), Agentic AI, LLM Evaluation, Fine Tuning

MLOps & Cloud: AWS, GCP, Azure, Docker, CI/CD (GitHub Actions), Prometheus, Grafana, Weights & Biases

Data & Web: PostgreSQL, SQLite, DuckDB/Parquet, FastAPI, React, Angular, Streamlit, D3.js, Plotly Dash, Three.js